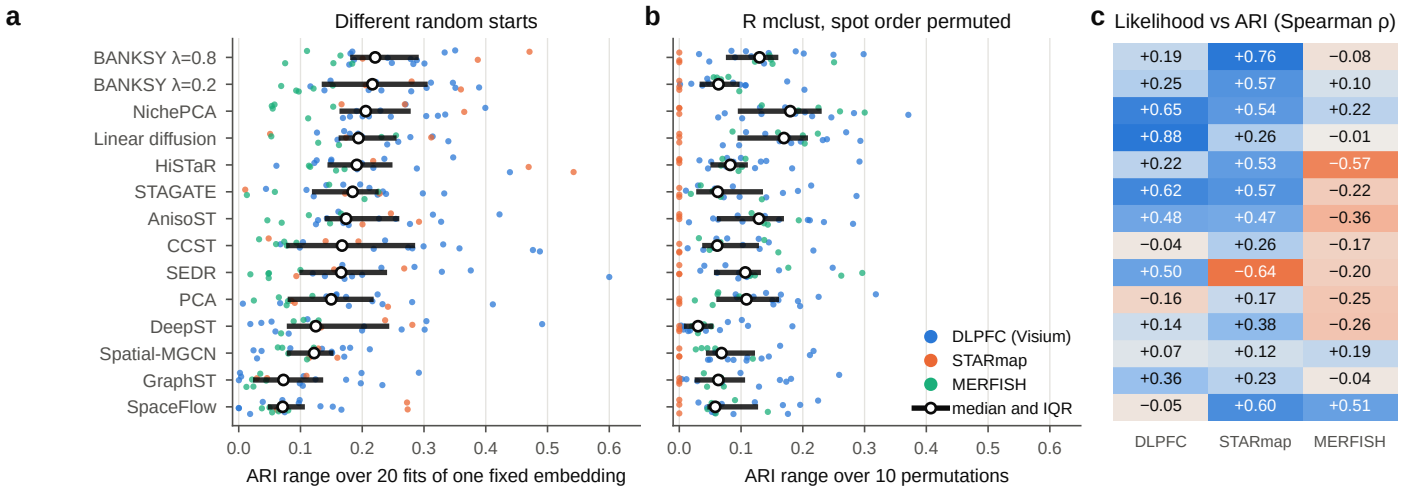

a
Different random starts
BANKSY λ=0.8
BANKSY λ=0.2
NichePCA
Linear diffusion
HiSTaR
STAGATE
AnisoST
CCST
SEDR
PCA
DeepST
Spatial-MGCN
GraphST
SpaceFlow
0.0
0.1
0.2
0.3
0.4
0.5
0.6
ARI range over 20 fits of one fixed embedding
b
R mclust, spot order permuted
DLPFC (Visium)
STARmap
MERFISH
median and IQR
ARI range over 10 permutations
c
Likelihood vs ARI (Spearman ρ)
+0.19 +0.76 −0.08
+0.25 +0.57 +0.10
+0.65 +0.54 +0.22
+0.88 +0.26 −0.01
+0.22 +0.53 −0.57
+0.62 +0.57 −0.22
+0.48 +0.47 −0.36
−0.04 +0.26 −0.17
+0.50 −0.64 −0.20
−0.16 +0.17 −0.25
+0.14 +0.38 −0.26
+0.07 +0.12 +0.19
+0.36 +0.23 −0.04
−0.05 +0.60 +0.51
DLPFC
STARmap
MERFISH